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# Introduction to Economics

## Class 5 — Competitive markets, demand and supply

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Universitat de Barcelona · Group A6

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### Slide 1 Title

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**Preview version.** The full slides and handout are posted after class.

Class 5 opens Block 2: markets. Block 1 asked what is possible and what it costs; from today the question is what sets the price. The first hour is the buyers' side, demand, built around a real price cut on the Barcelona metro; the second hour mirrors it on the sellers' side, supply. One curve at a time today, the two together on Thursday.

### Plan for today

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Block 1, one more look  
Block 2 begins: the metro's price cut  
A competitive market  
Demand: a curve, a move along it, a shift  
Break  
Project: the sign-up sheet  
Supply: a curve, a move along it, a shift  
Both curves on one picture  
Before Thursday

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### Slide 2 Plan for today

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The class at a glance. First a second look at two Block 1 ideas; then the metro's price cut, what a competitive market is, and demand: a curve, a move along it, a shift. After the break the project sign-up sheet, then supply built the same way, and finally both curves on one picture.

Two halves of fifty minutes with a ten-minute break between them.

### Block 1, one more look: same cost on both sides

**Pau bakes 4 croissants or 2 ensaïmadas an hour; Júlia 8 or 4. Which deal makes both better off?**

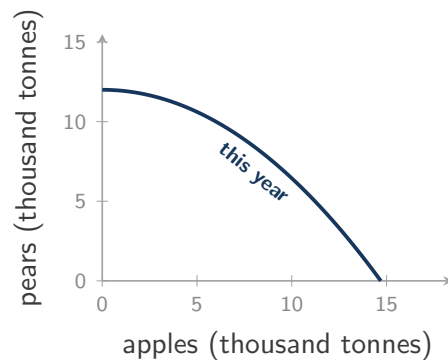
- A) Júlia bakes everything: she is faster at both
- B) Pau sells Júlia ensaïmadas at 1 croissant each
- C) Pau sells Júlia ensaïmadas at 3 croissants each
- D) None: an ensaïmada costs both 2 croissants

### Slide 3 Block 1, one more look: same cost on both sides

Pau bakes 4 croissants or 2 ensaïmadas an hour, Júlia 8 or 4. Before class, work out what one ensaïmada costs each of them in croissants, then decide which deal, if any, makes both better off, and why the others fail.

Revealed in class.

### Block 1, one more look: what moves the frontier?



- 1 A storm flattens half the pear orchards.
- 2 The world price of apples doubles.
- 3 The island grows more pears, fewer apples.
- 4 New irrigation helps both crops.

Two minutes. Which of the four move Thursday's frontier — and how?

### Slide 4 Block 1, one more look: what moves the frontier?

Thursday's island, apples and pears, and four events. Before class, decide for each one whether it moves the frontier, and how, or leaves it where it is. Think about what the frontier is made of, and about what a price can and cannot change.

Revealed in class.

## Block 2: markets

What sets the price: demand and supply

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### Slide 5 Block 2: markets

Block 2 begins. Block 1 asked what is possible and what it costs; Block 2 asks what sets the price, in the reading's model: a competitive market with a demand side and a supply side.

### September 2022: a month of metro for 20 euros, not 40



T-usual, one zone, from 1 September 2022

### What did people do?

ATM (Autoritat del Transport Metropolità), fares from 1 September 2022: the T-usual at half price.

### Slide 6 September 2022: a month of metro for 20 euros, not 40

The hook of the class. From 1 September 2022 the T-usual, the monthly pass for one zone, cost 20 euros instead of 40: a 30 % discount paid by the state in its energy-crisis package, topped up to 50 % by the Generalitat, the AMB and the city council; the half price stayed through 2023. One seller, and a price cut by decree. What did people do?

## What people did

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# Revealed in class

Before class: write down your guess — did journeys go up, and by roughly how much?

## Slide 7 What people did

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Revealed in class.

Beforehand: write down your guess. Did journeys go up after the price cut, and by roughly how much? Then ask whether a change in journeys could have causes other than the price, and what they might be.

## A competitive market: many buyers, many sellers, one price

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- One good, the same from every seller
- So many buyers and sellers that no one can move the price
- Everyone takes the price as given

Mercabarna's tomatoes: close. The metro: one seller — its buyers still follow a demand curve.

## Slide 8 A competitive market: many buyers, many sellers, one price

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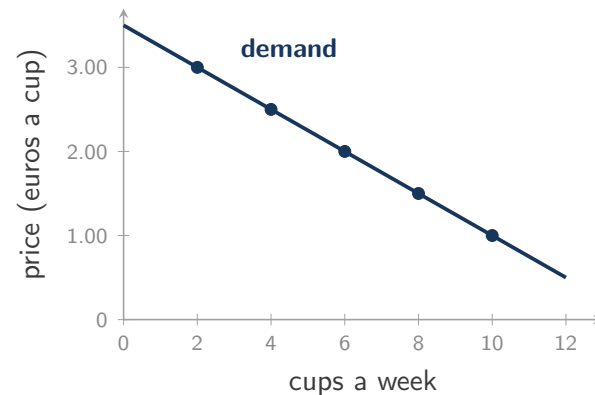
A market is the buyers and sellers of one good. A competitive market is one where the good is the same from every seller and there are so many buyers and sellers that none of them can move the price: everyone takes it as given. Tomatoes at Mercabarna, the wholesale market in the Zona Franca, come close.

The metro is not a competitive market: one seller, and a price set by decree. It opens the class because the buyers' side is where the model gets its first test: what happens to journeys when the price of the pass changes. The seller's side is where the metro differs, and we leave that alone.

## Demand: at each price, how many cups Mireia buys

| price (€ a cup) | cups a week |
|-----------------|-------------|
| 1.00            | 10          |
| 1.50            | 8           |
| 2.00            | 6           |
| 2.50            | 4           |
| 3.00            | 2           |

Mireia's demand schedule



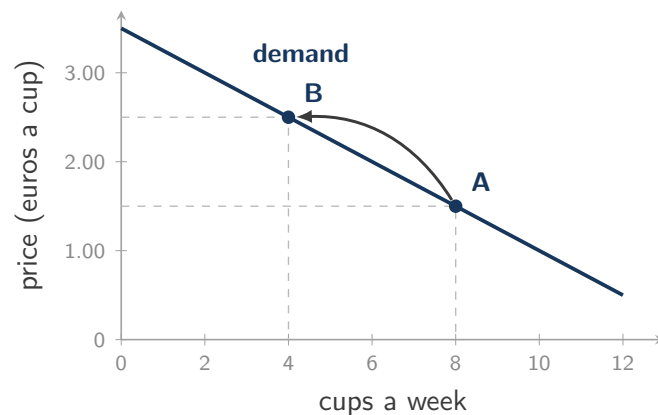
**Law of demand:** other things equal, the price rises, the quantity demanded falls.

## Slide 9 Demand: at each price, how many cups Mireia buys

One buyer. Mireia's demand schedule lists, for each price of a café amb llet at the faculty bar, how many cups a week she buys: 10 at 1 euro, 8 at 1.50, down to 2 at 3 euros. Plotted and joined, the points are her demand curve. Price goes on the vertical axis and quantity on the horizontal, by convention.

The quantity demanded is the amount buyers are willing and able to buy at a given price. The law of demand: other things equal, when the price rises the quantity demanded falls, so the curve slopes down. "Other things equal" is the phrase to underline: the schedule holds Mireia's income, her tastes and every other price fixed.

## A dearer cup: fewer cups, the same curve

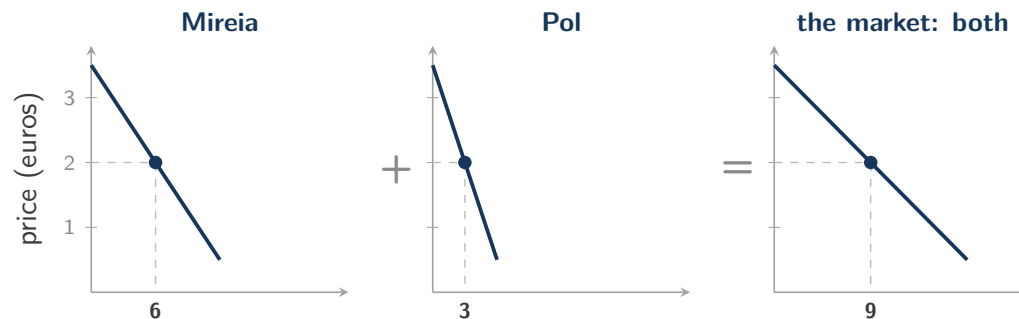


A to B: the quantity demanded fell from 8 to 4. The curve — demand — did not move.

## Slide 10 A dearer cup: fewer cups, the same curve

The bar raises the price from 1.50 to 2.50. Mireia moves from A to B along her curve: 8 cups a week become 4. Two sentences to keep: the quantity demanded fell; demand, the curve, did not change. A price change never moves the curve, because the curve already contains every price.

## Market demand: at each price, add the buyers up



At 2 euros, Mireia 6 + Pol 3 = 9 a week: the buyers' curves added sideways.

## Slide 11 Market demand: at each price, add the buyers up

The market is all the buyers. At each price, add up what each of them wants: at 2 euros Mireia wants six cups and Pol three, so the market wants nine, and the same at every other price. On the picture that is adding the curves sideways, horizontally, never vertically; the dashed guide at 2 euros shows it. In the reading's words, the market demand curve is the horizontal sum of the individual demand curves.

The market curve slopes down for the same reason as each buyer's, and for one more: at a lower price some people buy who bought nothing before.

## In pairs: add up the market

Three buyers of cups a week. Fill the last column; then the question.

| price | Ona | Biel | Roser | the market |
|-------|-----|------|-------|------------|
| 1.00  | 6   | 4    | 5     |            |
| 1.50  | 5   | 3    | 4     |            |
| 2.00  | 4   | 2    | 3     |            |
| 2.50  | 3   | 1    | 2     |            |

The bar raises the price from 1.50 to 2.00: a move along the market curve, or a shift of it?

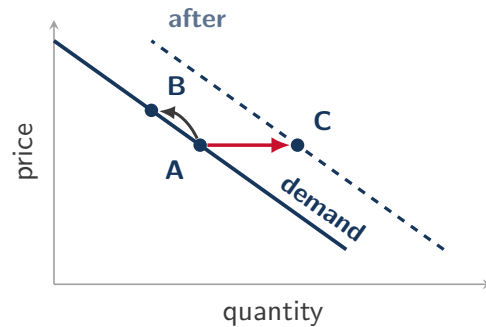
Three minutes.

## Slide 12 In pairs: add up the market

Three buyers' schedules. Before class, fill the market column: at each price, add up what the three of them want. Then decide whether a price rise from 1.50 to 2.00 is a move along the market curve or a shift of it, and say in one sentence what changed and what did not.

Revealed in class.

## Price moves you along the curve. All else moves the curve.



**A → B**

a move along the curve:  
the price changed

**A → C**

the curve moved:  
something else changed

Along the curve: the **quantity demanded** changes. A shift: **demand** changes.

**Slide 13** Price moves you along the curve. All else moves the curve.

The slide of the day. From A to B the price rose and we slid along the curve. From A to C, at the same price, the market wants more: the whole curve has moved to the right, which only happens when something other than the price changed.

Along the curve, the quantity demanded changes. A shift, and demand changes. A shift to the right means more wanted at every price; to the left, less at every price.

**Slide 14** What moves the demand curve

## What moves the demand curve

- **Income** rises: normal goods up, inferior goods down
- Price of a **substitute**: tea dearer, more coffee
- Price of a **complement**: croissants dearer, less coffee
- **Tastes** and **expectations**
- The **number of buyers**

Right: more wanted at every price. Left: less at every price.

What shifts the demand curve: anything that matters to buyers other than the good's own price. Income: when it rises, demand for a normal good shifts right and demand for an inferior good shifts left (the cheapest instant noodles, once you find a job). The price of a substitute, a good you buy instead: tea dearer, more coffee, a shift right. The price of a complement, a good you use together with it: croissants dearer, less coffee, a shift left.

Tastes, and advertising, which works on them. Expectations: if you expect the price to rise next month, you buy now. And the number of buyers: more people on the campus, more cups at every price.

## In pairs: demand for metro journeys — along the curve, or a shift?

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- 1 The T-usual drops from 40 to 20 euros.
- 2 Offices reopen after Covid.
- 3 Petrol gets dearer.
- 4 Taxis get cheaper.
- 5 Cycling to work becomes the fashion.

Three minutes. A move along the curve or a shift — and which way?

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## Slide 15 In pairs: demand for metro journeys

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Five events in the metro's market, with the curve showing journeys at each price of the pass. Before class, sort them: a move along the curve or a shift, and which way? The test from the previous slides: did the metro's own price change, or something else?

Revealed in class.

## Back to the metro

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# Revealed in class

From September 2022 the monthly pass, the T-usual, cost half its old price;  
in 2023 the ten-ride ticket, the T-casual, was back at its old price.

Before class: for each ticket, a move along its demand curve, or a shift of it?

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## Slide 16 Back to the metro

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Revealed in class.

Beforehand: from September 2022 the monthly pass, the T-usual, cost half its old price, 20 euros instead of 40, and it stayed there in 2023; the ten-ride ticket, the T-casual, was back at its old price of 11.35 euros in 2023. For each ticket, is what happened to its buyers a move along its demand curve, or a shift of the curve? Use the list of what moves a demand curve.

## Slide 17 Break

Ten minutes. Sit with your project group afterwards: the sign-up sheet for the idea meetings goes round right after the break.

# Break

Ten minutes

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## Project: sign up for your idea meeting

- The sheet goes round now: one slot per group, six minutes
- Thursday 8 October, right after class
- The day before your slot: your idea in one paragraph, by e-mail
- Full brief on the course page



course page

## Slide 18 Project: sign up for your idea meeting

The sign-up sheet for the idea meetings: one slot per group, six minutes each, on Thursday 8 October straight after class. The day before your slot, e-mail your idea in one paragraph, so the six minutes go on the question rather than on explaining it. The project brief on the course page (QR code) says what makes a good question: specific, local, something you noticed yourselves.

Groups were formed in Class 4; anyone still without one joins a group now. The topic is fixed on Thursday 15 October; the poster session is on Thursday 5 November.

# Supply

The sellers' side of the same market

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## Slide 19 Supply

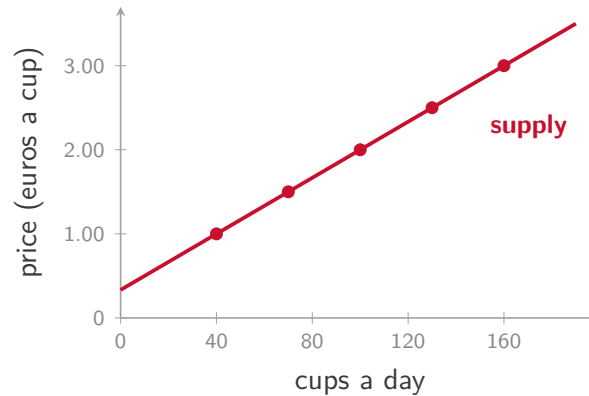
Supply: the sellers' side of the same market. The same three questions, mirrored: how much does one seller offer at each price, how do we add the sellers up, and what moves the curve.

### Supply: at each price, how many cups Bar Nou makes

| price (€ a cup) | cups a day |
|-----------------|------------|
| 1.00            | 40         |
| 1.50            | 70         |
| 2.00            | 100        |
| 2.50            | 130        |
| 3.00            | 160        |

Bar Nou's supply schedule

**Law of supply:** other things equal, the price rises, the quantity supplied rises.

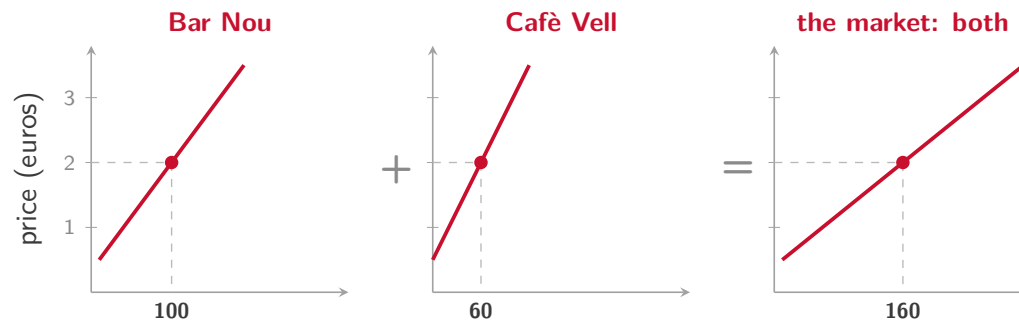


## Slide 20 Supply: at each price, how many cups Bar Nou makes

One seller. Bar Nou's supply schedule lists, for each price, how many cups of café amb llet a day it makes: 40 at 1 euro, 70 at 1.50, up to 160 at 3 euros. Joined, the points are its supply curve. The quantity supplied is the amount sellers are willing and able to sell at a given price. The law of supply: other things equal, when the price rises the quantity supplied rises, so the curve slopes up.

Why up: at a higher price it pays to open longer, put a second person on the machine and serve the slow customer too; each extra cup costs a little more to make, and a higher price covers it. Supply curves are red and demand curves navy throughout the course.

## Market supply: at each price, add the sellers up



At 2 euros, Bar Nou 100 + Cafè Vell 60 = 160 a day: the sellers' curves added sideways.

## Slide 21 Market supply: at each price, add the sellers up

The market is all the sellers. At each price, add up what each of them offers: at 2 euros Bar Nou makes 100 and Cafè Vell 60, so the market offers 160 cups a day. The same move as for the buyers: the curves are added sideways, never vertically. In the reading's words, the market supply curve is the horizontal sum of the individual supply curves.

Below 50 cents only Bar Nou sells, so the market curve has a kink there; the three panels are drawn from 50 cents up.

## In pairs: add up the sellers

Two sellers of cups a day. Fill the last column; then the two questions.

| price | Forn Gran | Bar Petit | the market |
|-------|-----------|-----------|------------|
| 1.00  | 30        | 10        |            |
| 1.50  | 50        | 20        |            |
| 2.00  | 70        | 30        |            |
| 2.50  | 90        | 40        |            |

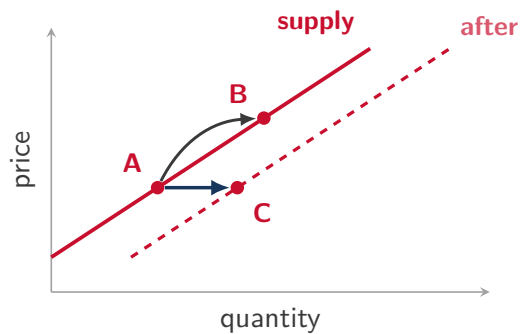
- 1 The price rises from 1.50 to 2.00 — along the curve or a shift?
- 2 Bar Petit closes — along the curve or a shift?

## Slide 22 In pairs: add up the sellers

Two sellers. Before class, fill the market column, then take the two questions: is each change a move along the market curve or a shift of it, and which way? Ask what changed in each case, the price or something else.

Revealed in class.

## Price moves you along the supply curve. All else moves the curve.



**A → B**

a move along the curve:  
the price changed

**A → C**

the curve moved:  
something else changed

Along the curve: the **quantity supplied** changes. A shift: **supply** changes.

**Slide 23** Price moves you along the supply curve. All else moves the curve.

The slide of the day, mirrored. From A to B the price rose and the seller slid up along the supply curve. From A to C, at the same price, the market offers more: the curve has moved to the right, because something other than the price changed.

Along the curve, the quantity supplied changes. A shift, and supply changes. Right means more offered at every price; left, less. "All else" is wider than costs: the number of sellers and expectations move supply too.

## What moves the supply curve

- **Input prices:** beans, milk, wages, rent
- **Technology:** more cups an hour
- **Expectations** about the price
- The **number of sellers**

Right: more offered at every price. Left: less at every price.

**Slide 24** What moves the supply curve

What shifts the supply curve: anything that matters to sellers other than the good's own price. Input prices, such as beans, milk, wages and rent: dearer inputs mean less offered at every price, a shift left. Technology, such as a machine that makes two cups a minute: a shift right. Expectations about the price: a seller who expects a higher price next month and can store the good holds back today, so today's supply shifts left; a firm that expects higher prices builds capacity, and later supply shifts right. The number of sellers: more bars on the street, a shift right.

The reading adds the weather and other natural factors, and how profitable the other goods a firm could make instead are.

## In pairs: the street's supply of coffee — along it, or a shift?

- 1 Coffee beans double in price.
- 2 A new machine makes two cups a minute.
- 3 The price of a cup rises to 2 euros.
- 4 Three new bars open on the street.
- 5 Baristas' wages rise.

Three minutes. A move along the curve or a shift — and which way?

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## Slide 25 In pairs: the street's supply of coffee

Five events on a street of cafès, with the curve showing the street's supply of coffee, all the cafès together. Before class, sort them: a move along the curve or a shift, and which way? For each one, ask whether it is the price of a cup that changed or something the sellers care about.

Revealed in class.

## Two pairs of words that are not the same

| what changed  | on the picture   | say                                                 |
|---------------|------------------|-----------------------------------------------------|
| the price     | along the curve  | <b>quantity demanded</b> · <b>quantity supplied</b> |
| anything else | the curve shifts | <b>demand</b> · <b>supply</b>                       |

"Demand rose" means the curve moved. "Quantity demanded rose" means the price fell.

## Slide 26 Two pairs of words that are not the same

The table says it. When the price changes, you move along the curve, and the words are quantity demanded or quantity supplied. When anything else changes, the curve shifts, and the words are demand or supply. "Anything else" means anything else that matters to buyers (for demand) or to sellers (for supply): an input price moves supply, not demand.

"Demand rose" means the curve moved. "Quantity demanded rose" means the price fell. Midterm 1 marks this wording.

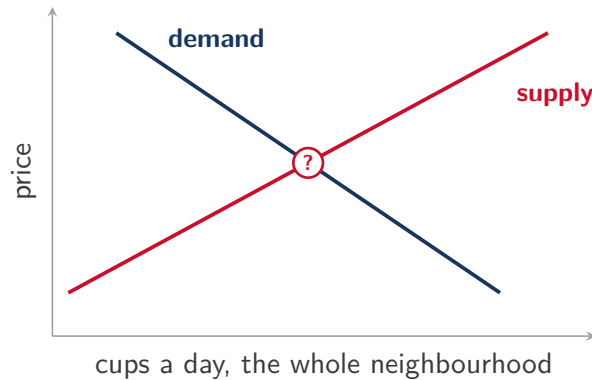
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## Demand and supply on one picture: where they cross, Thursday



Thursday: what happens at the crossing — and what moves it.

## Slide 27 Demand and supply on one picture

Both curves on one picture: the whole neighbourhood's buyers (navy) and sellers (red) of càfé amb llet. They cross once. What happens at the crossing, why the market goes there, and what happens to the price when one curve moves, is Thursday's class: every shift from today's two sorting exercises becomes a price change there.

## Price moves you along a curve. Everything else moves the curve.

Demand and supply: anything but the good's own price.

The frontier: resources and technology move it.

## Slide 28 In one line

The class in one line: price moves you along a curve; everything else moves the curve. For demand and supply, "everything else" is anything but the good's own price. The frontier of Block 1 has its own version of the rule: resources and technology move it, the island's choices move you along it, and prices are not on its axes.

## Before Thursday 8 October

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- Read Mankiw & Taylor, Ch. 3, pp. 44–52 — supply and demand together, prices as signals, changes in equilibrium
- Practise Block 1 in Sextant: the units for Classes 2 and 3
- Project: idea meetings Thursday, right after class; your idea by e-mail the day before your slot
- Thursday 8 October — equilibrium: where the two curves meet



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## Slide 29 Before Thursday 8 October

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Before Thursday: read Mankiw & Taylor chapter 3, pages 44 to 52, on supply and demand together, prices as signals and changes in equilibrium. Practise Block 1 in Sextant with the units for Classes 2 and 3. Project: the idea meetings are on Thursday straight after class; your idea in one paragraph by e-mail the day before your slot.

Thursday: equilibrium, where the two curves meet.

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